Frederick Wang

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Education

University of Toronto – M.Sc. Applied Computing (AI Concentration) **September 2026 – Present**
Sample Coursework: Neural Networks, Deep Learning, Imitation Learning Robotics, Computational Linguistics

University of British Columbia – B.Sc. Computer Science and Statistics **September 2021 – May 2026**
Awards: Dean's List (2021-2026), Trek Excellence Scholarship (2022), J. Fred Muir Scholarship (2022)
Sample Coursework: Machine Learning, Relational Databases, Data Structures and Algorithms, Software Engineering, Computer Hardware and Operating Systems, Statistical Inference, Stochastic Processes

Skills

Languages: Python, SQL, R, Java, C++, C, JavaScript
ML/AI: PyTorch, scikit-learn, Hugging Face, RAG, MCP, Imitation Learning, Time-Series Forecasting, Bayesian Modelling
Data & Cloud: Spark, Databricks, Unity Catalog, Great Expectations, AWS (Bedrock, OpenSearch, S3), Azure, SQL Server
Tools: Git, CARLA, Next.js, Node.js, JupyterLab

Work Experience

Data Engineer Intern – British Columbia Investment Management **May 2025 – December 2025**
- Secured 1st place in internal hackathon by engineering a schedule optimization algorithm (25% efficiency gain); led production deployment, projected to unlock $20M in enterprise value for a portfolio company
- Engineered and deployed production MCP server for Claude.ai to Unity Catalog integration via Databricks apps, achieving a 50% efficiency gain in company-wide ad hoc data analysis
- Added SQL Server Managed Instance support to open-source Great Expectations for Databricks integration
- Architected and developed a RAG knowledge base for natural language querying of unstructured data

Data Analyst Intern – Providence Research **January 2024 – August 2024**
- Leveraged machine learning to predict patient surgery outcomes based on risk profiles and surgery type
- Utilized R and RStudio to clean, analyze, and model ankle surgery outcome data for visualization and statistical testing

Projects

Recoverability-Aware World Models for Safe Autonomous Driving **October 2026 – Present**
- Co-developing a world-model training objective that preserves whether a vehicle can still recover after a decision, distilled into a DAgger-trained driving policy to reduce false "safe" judgments made by prediction-only world models
- Building a headless CARLA simulation-mining pipeline to automatically discover unrecoverable driving scenes, with NVIDIA Cosmos generating counterfactual variants for recovery-label training

Undercurrent: Foundation Model Commodity Forecasting **September 2026 – Present**
- Benchmarking Chronos-2 with native covariates against a hierarchical Bayesian driver model combined via Bayesian Predictive Synthesis, scoring leakage-safe commodity price forecasts on CRPS and calibration

GreenLife – AWS x UBC CIC Generative AI and Sustainability Hackathon – 1st Place **October 2024**
- Engineered a full-stack Next.js platform to democratize access to personalized Canadian sustainability incentives, integrating a Python-based RAG pipeline powered by AWS Bedrock, OpenSearch, and S3

Teaching Experience

Teaching Assistant – UofT Department of Computer Science **September 2026 – Present**
- Lead weekly tutorials and labs for CSC207, teaching Java programming, object-oriented software design, Git-based collaboration, and effective project teamwork in a software-development setting

Teaching Assistant – UBC Departments of Computer Science and Statistics **September 2024 – April 2026**
- Mentored 100+ students on data science topics through hosting office hours and providing data science project guidance and evaluation
- Led weekly tutorials on regression and classification machine learning models in R on JupyterLab
- Facilitated lectures in a 400-level advanced relational databases course, covering query processing and optimization, transaction management, concurrency control, and crash recovery